

Visualization Analysis of Spatiotemporal Data of COVID-19

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Abstract—Aiming at the problem of COVID-19 epidemic data visualization, this paper proposes a spatiotemporal visualization analysis method based on the technology of scraping crawler and visualization, and carries on the visualization analysis and research, intuitively shows the development and change of the epidemic situation in different countries and regions, and excavates its spatiotemporal variation rules. Firstly, we use scrapy crawler framework to collect COVID-19 epidemic data; then, the collected data were cleaned and processed to construct a spatiotemporal data set of COVID-19 epidemic; finally, pyecharts is used to analyze the dataset data visually. The results showed the changes and trends of epidemic situation in different countries and regions, and provided reference for epidemic prevention and control.

Keywords—COVID-19, Visual analysis, Scrapy crawler framework, Spatiotemporal data set component

I. INTRODUCTION

Since the end of 2019, COVID-19 has spread all over the world. As of June 2021, nearly 17.7 million people have been diagnosed with infection, resulting in more than 3.8 million deaths [1]. COVID-19 is highly infectious and has a wide range of influence, which has attracted the attention of the whole world. The world's major mainstream websites update the latest epidemic data in real time to provide better support for epidemic prevention and control. However, simple digital changes make the data display of the epidemic not intuitive enough, so the visualization of COVID-19 epidemic data is particularly important. The visualization of COVID-19 epidemic data can directly display the development and change trend of the epidemic, help to excavate its spatiotemporal change law, provide reference for the party and government organs and enterprises to scientifically predict the trend of the epidemic, and provide scientific basis for different countries and regions to improve the epidemic prevention and control.

This paper provides a method of spatiotemporal visualization analysis of COVID-19 epidemic situation, and analyzes its spatiotemporal variation law from the perspectives of spatiotemporal visualization analysis, time series visualization analysis and comparative analysis. The method flow chart of this paper is shown in Fig. 1: first of all, this paper takes Tencent real-time epidemic tracking website and Baidu epidemic map as data sources, crawls epidemic data through the scrapercrawler framework, and stores the data in MySQL database; then, the data were cleaned and processed, and the spatiotemporal data set of COVID-19 epidemic was constructed by data aggregation; finally, through the spatiotemporal visualization analysis method, the data were visualized to show the development and changes of epidemic

situation in different countries and regions, and provide reference for epidemic prevention and control.

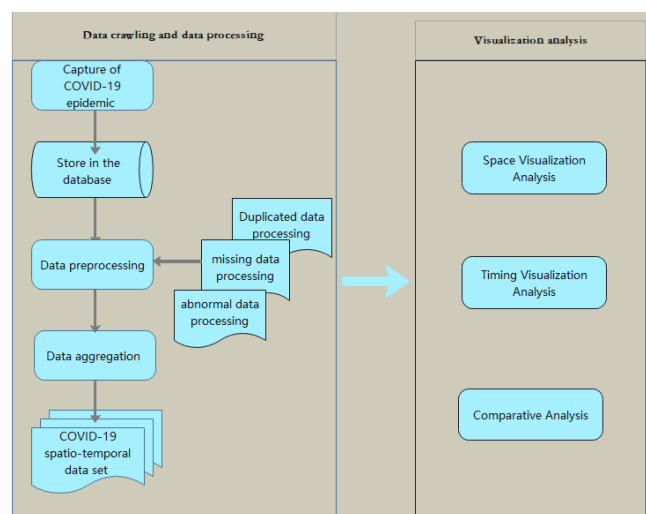


Fig. 1. Flow chart of the method

II. RELATED WORK

COVID-19 epidemic has the characteristics of rapid transmission, wide spread and difficult prevention and control. It has become a global pandemic and a major event in the field of public health. Nowadays, visualization technology is widely used in the field of infectious diseases and health, focusing on spatiotemporal pattern analysis of infectious diseases, multi-dimensional association analysis and detection and early warning. For example: Jiangping Huang et al [2] established a global epidemic prediction system by combining epidemic prediction model with global epidemic data, which helps us understand and reduce the impact of epidemic situation and future pandemic, and provides valuable reference for policy makers. With the help of GIS and big data technology, Chenghu Zhou et al [3] analyzed the spatial characteristics of disease, material, population and social psychology from the three dimensions of individual, group and region, quickly responded to and analyzed the epidemic situation, understood the development law of the epidemic situation, and provided timely support for the decision-making and action of epidemic prevention and control. Based on the confirmed infection cases and location information, Zhenxin Fu et al [4] established a risk area early warning system to warn people to stay away from such hot spots and reduce the risk of spreading diseases through droplet or contact. Xingshu Chen et al [5], through public opinion analysis on topics related to "COVID-19 epidemic", visualized the spatiotemporal evolution process of network public opinion in

this epidemic event, and showed three stages of netizens' attitude towards the pneumonia epidemic through text clustering and emotion analysis. Bingchuan Jiang et al [6] put forward a method to analyze the epidemic situation of COVID-19 by using geographic knowledge map combined with interactive visualization. Through multi view collaborative interaction such as map distribution visualization, map visualization and trajectory visualization, the epidemic situation of COVID-19 was analyzed and experimented, which provided a new idea and method for epidemic situation analysis. Jianxiang Liu et al. [7] took the epidemic data of Henan Province as an example, showed the epidemic situation of COVID-19, mined the source characteristics of COVID-19 epidemic, summarized the transmission mode of COVID-19 epidemic, predicted the future trend of COVID-19 epidemic, and provided scientific basis for the prevention and control of COVID-19 epidemic in Henan Province. Sichen Jin et al [8] proposed a visualization analysis system of infectious diseases based on multi-dimensional spatio-temporal data, which can intuitively analyze the spatio-temporal patterns of infectious diseases, interactively mine the association and similarity between different diseases and regions, and effectively help users mine the spatio-temporal patterns of infectious diseases, so as to better prevent, control and analyze infectious diseases. According to the characteristics of infectious disease transmission data, Xiaohui Chen et al [9] summarized the visualization methods of infectious disease transmission data according to four categories: map based, space-time cube based, time-series based and relationship based, which provided a new perspective for infectious disease transmission analysis. In general, a lot of work has been done on the visualization of COVID-19 epidemic data at home and abroad. There are many research results on the visualization and visualization analysis of COVID-19 epidemic data, but there are also many problems in the existing research results. The existing research mainly focuses on the local epidemic situation and epidemic prediction, and ignores the global visual analysis of COVID-19.

The contribution of this paper is to apply Scrapy crawler technology, data processing technology and visualization technology to the visualization analysis of COVID-19 epidemic data, analyze the spatiotemporal data of COVID-19 epidemic on a global scale, display the development and change of COVID-19 epidemic through spatiotemporal visualization analysis, and mine its internal laws to provide scientific basis for epidemic prevention and control.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Hardware and Software Environment

The experimental software and hardware environment selected in this paper is shown in Table I. Ubuntu system is an open source Linux operating system, which has an efficient and stable file management system, generates less system garbage, and has a powerful command line that can execute all operating instructions. Scrapy crawler framework is an asynchronous processing framework based on Twisted, which is implemented in pure Python language. It has clear architecture, low coupling between modules, strong scalability, and can flexibly complete the crawling of COVID-19 epidemic data. pycharm is chosen as the development tool to crawl epidemic data because pycharm is the most widely used integrated development environment of Python programming language, which supports code analysis and

coding assistance, and supports the integrity of Python ecology. As an open source multi-user and multi thread database server, MySQL has the advantages of small size, fast running speed and easy to use. As the core data analysis support library of python, pandas is a powerful and flexible open source data analysis tool with fast, flexible and clear data structure, which aims to process relational and marked data simply, intuitively and quickly. In this paper, Pyecharts is chosen as a data visualization tool because of its simple API design, smooth use, support for mainstream notebook environment, including 30+ common charts, and support for 400+ map files, which provides strong support for data visualization.

TABLE I. EXPERIMENTAL SOFTWARE AND HARDWARE ENVIRONMENT

Experimental Environment	Environmental Configuration
System environments	Ubuntu20.04
Crawler frame	Scrapy
Development tool	Pycharm
Data base	MySQL
Data processing tools	Pandas
Visualization tool	Pyecharts

B. Data Acquisition

COVID-19 epidemic has gathered the attention of the whole world, and a large number of real-time reporting websites of COVID-19 epidemic have emerged, such as Tencent epidemic map, Baidu epidemic map, Johns Hopkins University epidemic website, etc. This paper selects Tencent epidemic map as the data source, and uses Scrapy open source crawler framework [13] Based on Python language to collect epidemic data. The architecture of Scrapy crawler framework [11] is shown in Fig. 2. Its main components are scrapy engine, scheduler, item pipeline, downloader and spiders. Scrapy engine is responsible for processing the data stream of the whole system and triggering the real object, which is the core of the whole framework. The scheduler accepts the request from the engine and adds it to the queue. When the engine requests again, it provides the request to the engine. The downloader downloads the content of the web page and returns it to spiders. Spiders defines the crawling logic and the parsing rules of web pages, which is mainly responsible for parsing the response and generating the extraction results and new requests. Item pipeline is responsible for processing items extracted from web pages by spiders, cleaning, verifying and storing data.

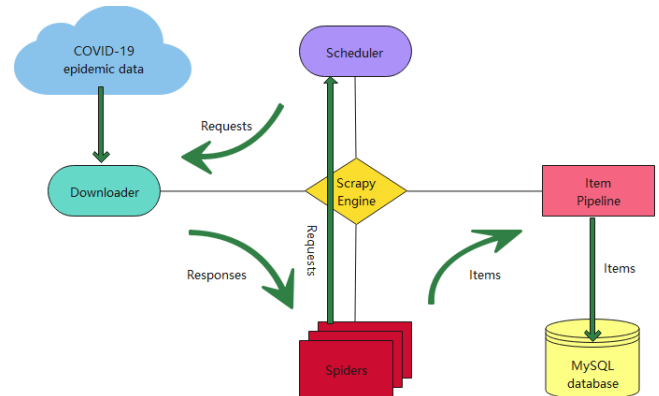


Fig. 2. Scrapy crawler framework

The crawling process of COVID-19 epidemic data is as follows:

- Engine first opens the Tencent epidemic map website, finds the spider that handles the website, and requests a URL to be crawled from the spider.
- The engine obtains the first URL to crawl from spider and schedules it in the form of request through scheduler.
- The engine requests the next URL to crawl from the scheduler, returns the next URL to crawl to the engine, and the engine forwards the URL to the downloader for download.
- After downloading, the downloader will generate the response of the page and send it to the engine.
- The engine receives the response from the downloader and sends it to spider for processing.
- After processing the response, spider returns the extracted item and new request to engine.
- The engine returns the item returned by spider to the item pipeline and the new URL to the scheduler.
- Repeat steps 2-7 until there are no more requests in the scheduler and the engine stops crawling.

In order to improve the efficiency of crawler crawling, this paper designs a multi pipeline crawler parsing strategy to deal with multiple crawlers crawling epidemic data, which is more efficient than a single pipeline parsing crawling data. Because the data crawled by the crawler is stored in the project pipeline by default every time, it will occupy a large amount of memory every time. In order to better store the crawled epidemic data, we decided to use MySQL database to store the epidemic data. In order to connect to MySQL database persistently and with low consumption, this paper uses pymysql, a third-party library in Python, to connect with MySQL. By defining a database connection function, it can call MySQL database in real time to store and process the epidemic data in the next step.

C. Data Processing

COVID-19 epidemic data includes time, region, longitude and latitude information, cumulative confirmed cases, existing confirmed cases, cured cases, deaths, daily new confirmed cases, input confirmed cases and other fields. Due to the different time of No. 0 case in different countries and different statistical data methods among different countries, there are chaotic, repetitive and incomplete data in COVID-19 epidemic data. If these data are not directly applied without data cleaning, it will directly or indirectly affect the accuracy of visualization results of epidemic data. So we must go through data cleaning in order to better carry out data processing and analysis. In this paper, python data analysis package pandas is used to clean the epidemic data, including missing value processing, duplicate value processing and abnormal value processing. First, read through pandas read_sql() method reads the epidemic data stored in MySQL database; Secondly, the duplicate epidemic data were deleted by dropna() method; Then, fill in the null data with fillna() method; Finally, the missing data is added through the append() method.

After cleaning and processing the epidemic data, in order to better visualize the epidemic data, the study decided to extract multiple fields of the epidemic data according to the characteristics of the epidemic data at home and abroad to build the domestic and foreign epidemic data sets, as shown in Table II and Table III.

TABLE II. FOREIGN EPIDEMIC DATASET

Field Name	Illustration
Country	Country
Continent	Continent
Date	Date
Confirm	Cumulative confirmed cases
Nowconfirm	Existing confirmed cases
ConfirmAdd	New cases every day
Suspect	Suspect cases
Dead	Dead cases
Heal	Heal cases
Healrate	Cure rate
Deadrate	mortality rate

TABLE III. DOMESTIC EPIDEMIC DATASET

Field Name	Illustration
Province	Province
Latitude	Latitude
Longitude	Longitude
Date	Date
Confirm	Cumulative confirmed cases
Nowconfirm	Existing confirmed cases
ConfirmAdd	New cases every day
Suspect	Suspect cases
Dead	Dead cases
Heal	Heal cases
Healrate	Cure rate
Deadrate	mortality rate

D. Visualized Analysis

The visualization of COVID-19 epidemic data will show different changes with the changes of seasons and spatial changes. In order to better show the changes of the epidemic, the research decided to take the spatial and temporal changes of the epidemic data as the theme, and adopt the visualization method based on map [12, 13] and time sequence [14] to carry out visual analysis.

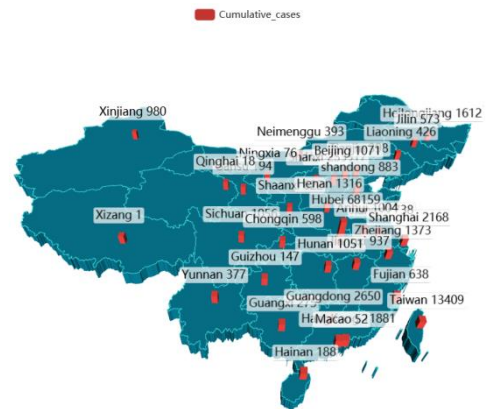
1) *Map-based Visualization Analysis*: Considering the spatial distribution characteristics of COVID-19 epidemic data, the study decided to use the map based visualization analysis method to analyze the epidemic data. Firstly, the epidemic data sets in Table II and Table III are used to extract the fields needed in the data sets; then, in order to make the effect of map based visualization better, we need to choose the appropriate RGB color according to the number of confirmed cases; finally, Pyecharts is used to analyze the epidemic data. As shown in Fig. 3 to Fig. 8, the COVID-19 epidemic data are based on the results of spatial distribution.

2) *Timing-based Visualization Analysis*: Based on the timing visualization analysis, the time change trend of the COVID-19 epidemic can be shown. Design reasonable time sequence visualization analysis diagram requires long enough time range and appropriate time units. This paper selects the 2020 COVID-19 epidemic data for visual analysis of epidemic data every day and monthly. As shown in Fig. 9 directly shows the development and changes of domestic epidemic in China. Fig. 10 shows the global month.

Comparative Analysis: The performance of COVID-19 epidemic data in different time periods and different spaces is very different. The study decided to use the method of comparative analysis to visually analyze the epidemic data. The data of the same time period in different countries and different time periods in the same country were visually analyzed to show the development and changes of the epidemic situation in different regions. Fig. 11 shows the word cloud chart of confirmed cases in many countries, Fig. 12 shows the bar chart of the top ten confirmed cases in the world, and shows the severity of the epidemic by comparing the bar charts of many countries. Fig. 13 shows the countries with more serious epidemic in the form of funnel chart.

E. Conclusion

The visualization analysis of COVID-19 epidemic situation will change with the change of time and space. This paper analyzes the development and change of COVID-19 epidemic situation from the perspectives of spatial visualization analysis, time series visualization analysis and comparative analysis.



epidemic prevention and control policy and the vaccination of COVID-19 vaccine, the COVID-19 epidemic has been effectively controlled, but sporadic confirmed cases will appear in some areas from time to time, such as Guangdong Province and Taiwan Province. Now it is still the key area of epidemic prevention and control in China. Combined with Fig. 7 and Fig. 8, the COVID-19 epidemic has spread to every corner of the world. Europe, South Asia, North America and South America have become the most severely affected areas. In particular, the number of confirmed patients in the United States, India and Brazil has exceeded 10 million, and the

number of confirmed patients in Europe has also exceeded 1 million. Through the spatial visualization analysis, we can clearly see which countries or a certain region of the country are more densely diagnosed patients, so that we can more intuitively judge the severity of the epidemic.

2) *Timing Visualization Analysis*: The visualization analysis of COVID-19 epidemic time series takes time change as the main line, and selects the epidemic data of China in 2020 and the top ten countries in the world as examples for visualization analysis, as shown in Fig. 9 and Fig.10.

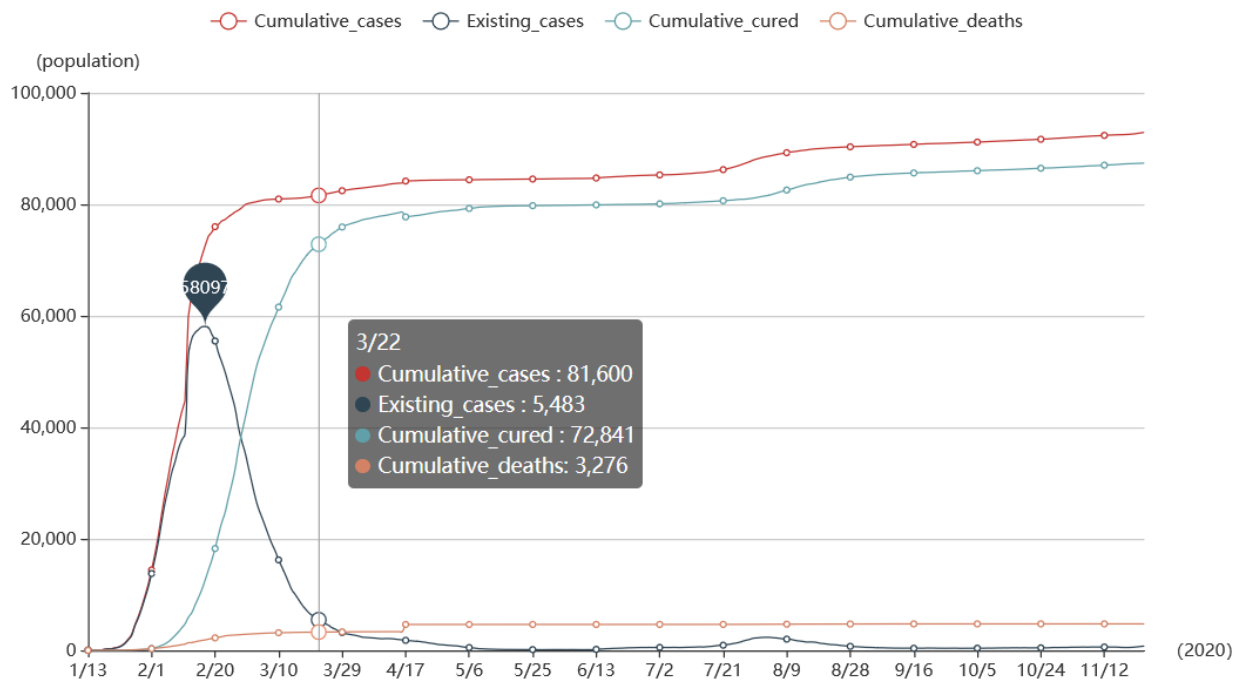


Fig. 9. Trend of Epidemic in China

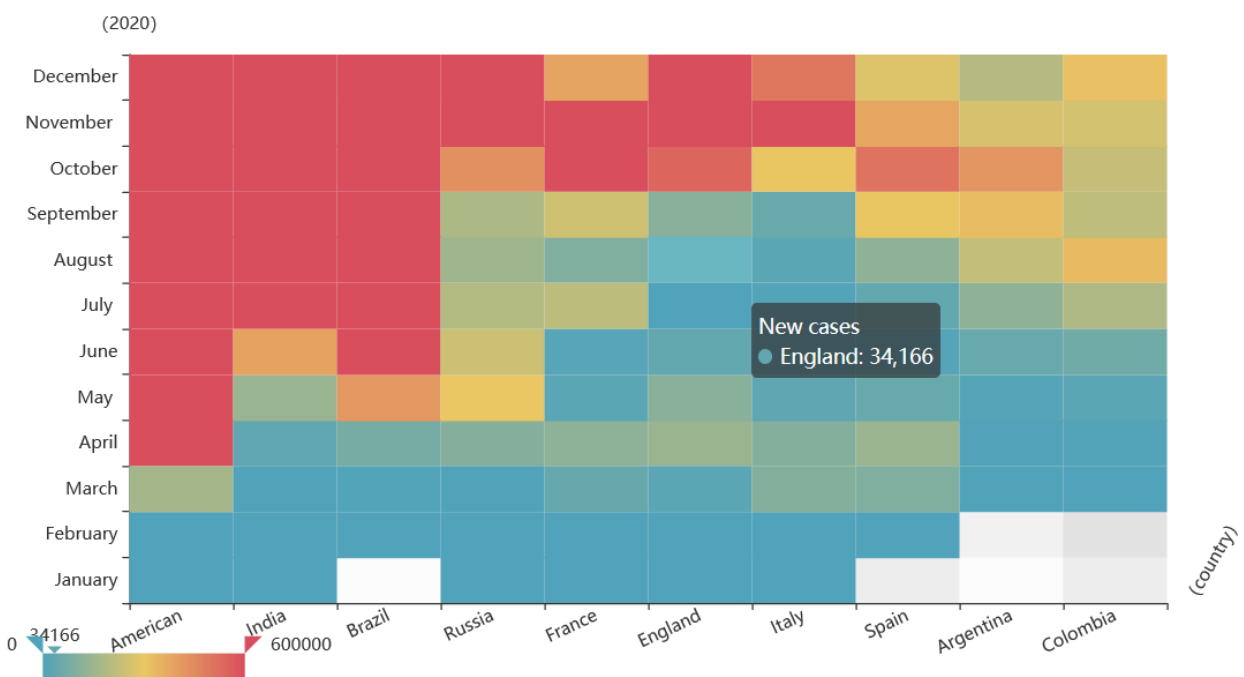


Fig. 10. Monthly new heat map in more countries

Fig. 10 shows the monthly heat map of newly diagnosed cases in the top ten countries in the world. Due to the large base of epidemic data, the study decided to make a visual analysis of the newly diagnosed data on a monthly basis.

3) *Comparative Analysis*: The COVID-19 epidemic has spread to all parts of the world. How to judge the severity of a country's epidemic is an urgent problem. The study decided to use comparative analysis to analyze the severity of a country's epidemic, as shown in Fig. 11 to Fig. 13.

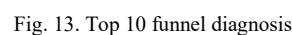
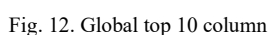
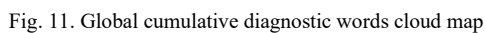


Fig. 11 shows the global cumulative number of confirmed word clouds. The size of the word cloud represents the number of confirmed cases in each country. The larger the word cloud is, the more serious the epidemic situation is in the corresponding country. In this paper, we use multi country confirmed data for comparative analysis. From the Fig. 12, we can clearly see that the three countries with the most serious epidemic situation in the world are the United States, India and Brazil. Fig. 12 and Fig. 13 respectively show the cumulative confirmed data of the top ten countries in the world by histogram and funnel chart, and show the serious situation of the epidemic situation by comparing the confirmed data of many countries.

IV. CONCLUSION

This paper proposes a method of COVID-19 epidemic spatiotemporal visualization analysis, which combines reptile technology and visualization technology to analyze the spatiotemporal variation of COVID-19 epidemic, and uses different visualization methods to analyze and show the development and change of COVID-19 epidemic. At the same time, combined with spatial visualization and time-series visualization, we analyzed the three stages of the development and change of the epidemic situation in China and in many countries. The analysis results can provide reference for the prevention and control of the epidemic situation in different countries. Future research will further combine data mining technology and visualization technology to deal with massive and high latitude epidemic data, and more effectively and deeply explore the spatiotemporal variation of the epidemic.

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